

1 Vacuum Suction System for End Effector

1. Purpose

A vacuum suction cup is integrated into the end effector to handle smooth, flat objects such as the compact disc (CD). The CD has a mass of 19 g, making it lightweight but difficult to grip securely using mechanical fingers alone. The suction cup provides a reliable and simple gripping method for such objects.

2. Operating Principle

The vacuum suction cup operates using **differential pressure**. When air is removed from the space between the suction cup and the object surface, atmospheric pressure acting externally becomes greater than the pressure inside the cup. This pressure difference creates a holding force.

The holding force is given by:

$$F = \Delta P \times A$$

Where:

- F = holding force
- ΔP = pressure difference
- A = effective suction area

The holding force increases as either the pressure difference or the suction area increases.

3. Required Holding Force Calculation

Mass of CD:

$$m = 0.019 \text{ kg}$$

Weight:

$$W = mg = 0.019 \times 9.81 \approx 0.19 \text{ N}$$

Using a safety factor of 3:

$$F_{\text{required}} \approx 0.57 \text{ N}$$

This required force is very small. Even a 30–40 mm diameter suction cup can easily provide a holding force significantly greater than this value under moderate vacuum conditions.

4. Required Vacuum System Components

To implement the suction gripping system, the following components are required:

1. **Flat Vacuum Suction Cup (30–40 mm diameter)**

The suction cup makes direct contact with the CD surface and creates the sealing interface necessary for vacuum formation.

2. **Mini DC Vacuum Pump (Vacuum Generator)**

The pump removes air from the space between the suction cup and the CD surface, creating the pressure difference required for holding force.

3. **Flexible Vacuum Hose**

Connects the suction cup to the vacuum pump and allows air to be evacuated efficiently.

4. **Electrical Switching Device (MOSFET or Relay Module)**

Used to safely control the vacuum pump from the microcontroller.

Mounting Adapter / Threaded Insert

Allows secure mechanical attachment of the suction cup to the gripper body.

These components together form a complete vacuum gripping subsystem without requiring compressed air or complex pneumatic valves.

5. Selected Suction Cup

A **flat** vacuum suction cup with a diameter of approximately **30–40 mm** is selected.

Flat cups are preferred over bellows or oval cups because:

- The CD has a **flat, smooth, non-porous** surface.
- Flat cups give **good stability** and a reliable seal on such surfaces.
- Bellows cups are mainly needed for uneven or flexible surfaces, which is not the case here.
- Oval cups are more suitable for long, narrow parts (e.g., strips or sheets), not circular CDs.

The diameter range (30–40 mm) is chosen for **practical** rather than theoretical reasons.

From the holding force formula $F = \Delta P \times A$, even a much smaller cup could theoretically hold a 19 g CD, but in practice a larger cup:

- Is more tolerant to small misalignments.
- Provides a wider sealing area.
- Is easier to mount mechanically to the gripper.

A soft **silicone or rubber** material is chosen to improve friction and sealing on the plastic CD surface, while avoiding scratches.

6. Vacuum Generation Method

Two main options were considered for generating vacuum:

1. **Pneumatic venturi vacuum ejector** driven by compressed air.
2. **Small DC electric vacuum pump.**

A venturi ejector would require:

- A compressed air supply (compressor),
- Additional pneumatic valves,
- Extra hoses and fittings,

which would increase cost and complexity for a small student prototype.

Therefore, a **mini DC vacuum pump** is selected as the vacuum generator. It:

- Operates directly from a low-voltage DC supply (e.g., 12 V),
- Connects to the suction cup via a simple flexible hose,
- Does not require an external compressor,
- Still implements **pneumatic fluid power**, since it uses air pressure difference to create holding force.

This choice keeps the system compact and easy to integrate while still demonstrating fluid power principles.

7. Control Method

The vacuum pump is controlled electronically by the microcontroller as a simple **ON/OFF actuator**:

- **Pump ON:** air is evacuated from the suction cup → vacuum is created → the CD is held.
- **Pump OFF:** air re-enters → vacuum is lost → the CD is released.

A digital output pin from the controller drives a **MOSFET or relay module**, which switches the higher-current pump safely. This avoids overloading the microcontroller pins and provides electrical isolation.

Variable vacuum levels are not required for this application because:

- The objects (CD, light parts) are very light,
- A single vacuum level is sufficient,
- The control logic remains simple and reliable.

This makes the suction system easy to integrate into the robot's motion sequence (e.g., “move above CD → pump ON → lift → move → pump OFF”).

8. System Justification

The final end-effector design is a **hybrid gripper**:

- **Mechanical fingers** handle irregular and fragile objects (egg, ball, screw, bolt).
- The **vacuum suction cup** handles smooth, flat objects (CD) where mechanical gripping is less reliable.

This division of roles lets each gripping method be used where it works best, improving overall reliability without overcomplicating the mechanism.

From a design point of view, the suction system:

- Demonstrates **pneumatic fluid power** using vacuum and differential pressure,
- Remains simple (one cup, one pump, one hose, one switch),
- Fits well within the constraints of a student prototype in terms of cost, size, and control complexity.